

TECHNICAL RULES

Monaco Energy Class

The Yacht Club de Monaco welcomes all teams interested in participating in the Monaco Energy Boat Challenge 2027.

The Monaco Energy Boat Challenge is owned and managed by the Yacht Club de Monaco.

The Monaco Energy Class was launched in 2018 and has seen a rise in interest from teams worldwide.

The participation of a team in Monaco is subject to a validation by a Technical Committee.

The Yacht Club de Monaco provides each selected team with a hull. Teams are responsible for designing and building the cockpit and propulsion system in accordance with the rules set out below.

This Technical Rule sets out the rules of the Monaco Energy Class and may be subject to modifications which, if necessary, will be communicated on the website of the Monaco Energy Boat Challenge. <https://monacoenergyboatchallenge.com/>

DOCUMENT CONTROL

Version 2027.1 - Issued 07/09/2026

by Yacht Club of Monaco

Table of Contents

TABLE OF CONTENTS	2
1. GENERAL.....	3
2. TERMS AND DEFINITIONS.....	4
3. CHANGES HISTORY	5
5. REQUIREMENTS	6
5.1. The Hull.....	6
5.2. Energy source and energy conversion	7
Flywheel energy storage	8
Heat energy storage	9
Gas storage.....	9
Batteries	10
Solar panels	11
Sails.....	11
5.3. The Engine, Propulsion & Steering	11
5.4. The Cockpit	12
Bulkhead.....	13
5.5. Electrics/Electronics	14
5.6. Monitoring.....	15
Internal	15
External	15
5.7. Kill cord & Emergency stop push button	18
Kill cord.....	18
Emergency stop push button	18
5.8. Safety & security Equipment	20
Safety equipment	20
Extinguisher.....	21
Hydrogen safety	22
Hydrogen Piping And Storage	22
Hydrogen Inside Enclosed Compartments	23
5.9. Pilot Requirements and Communication	25
Pilot requirements.....	Error! Bookmark not defined.
Communication	25
5.10. Appearance of the Boats.....	26
6. INSPECTIONS / RACING TEST	28
7. FAILURE & MODIFICATIONS	29
ANNEX I – HULL PACKAGE	30
ANNEX II - HULL DRAWINGS.....	31
ANNEX III – EXTERNAL MONITORING POWER SUPPLY INTERFACE.	32

1. General

- 1.1. The Monaco Energy Boat Challenge is owned and managed by the Yacht Club de Monaco.
- 1.2. The Yacht Club de Monaco will provide for the race a set of hulls to each selected team available in Monaco.
- 1.3. Teams will have to build a cockpit including the propulsion system with controls & steering and fit it to the hull.
- 1.4. The rules are published on the YCM's website and are binding for all participants and teams.
- 1.5. The Technical Regulations presented in this document serve as directives for the races. Situations that are not covered by the Rules will not be authorised.
- 1.6. This Technical Rule sets out the rules of the Monaco Energy Class and may be subject to modifications which, if necessary, will be communicated on the website of the Monaco Energy Boat Challenge. www.energyboatchallenge.com

2. Terms and definitions

Term	Definition
<i>Bulkhead</i>	upright partition separating the driver's compartment from the energy compartment
<i>Cockpit</i>	the pieces added to the hulls to build the boat
<i>Dead man's switch</i>	device that is designed to cut the power supply to the engine as soon as the Pilot loses control of the boat or when the Pilot leaves the vessel, whether voluntarily or involuntarily
<i>Emergency power button</i>	switch that, in case of emergency, can be used to disconnect the electrical power between energy sources and energy consuming devices
<i>Flywheel</i>	rotating part that is used to stored kinetic energy
<i>Flywheel housing</i>	non-moving part that is used to separate moving parts of a flywheel system to the non-moving parts
<i>Flywheel system</i>	system that stores kinetic energy and converts it into electrical power
<i>Fuel cell</i>	device that converts hydrogen into electrical power
<i>Heat storage medium</i>	material that stores energy thanks to a variation of temperature
<i>Hydrogen line</i>	pipes and devices that allow the hydrogen to go from the tank to the fuel cell
<i>LFL</i>	Lower Flammability Level
<i>Pilot</i>	team member who is qualified in accordance with the rules to Pilot the vessel
<i>Potential source of leakage</i>	any pipe fitting, including components interfaces (valves, filters, pressure regulators, and so on)
<i>Primary energy source</i>	stored energy in the boat before the beginning of the race.
<i>Secondary energy source</i>	energy produced during the race
<i>Sensible electrical component</i>	batteries, converters, MPPT, BMS, ECU
<i>Telemetry</i>	the process of recording and transmitting the readings of an instrument
<i>Waste</i>	any organic waste coming from the MEBC during the Event

3. Changes History

REQ Num	Change
REQ_2	Modified: Clarification on permitted hull design and alteration.
REQ_7	Modified: Precision regarding non-compliance.
REQ_48	Modified: Precision on the hulls' weight.
REQ_65	Deleted: (not applicable in 2027).
REQ_67	Modified: Merged with REQ_65.
REQ_93	Added: Clarification on hot surface.
REQ_153	Deleted: (not applicable in 2027).
REQ_155	Modified: Precision on nested containers.
REQ_173	Added: Precision on battery chemistry.
REQ_174	Deleted: (not applicable in 2027).
REQ_175	Deleted: (not applicable in 2027).
REQ_176	Deleted: (not applicable in 2027).
REQ_178	Deleted: (not applicable in 2027).
REQ_179	Deleted: (not applicable in 2027).
REQ_180	Deleted: (not applicable in 2027).
REQ_181	Modified: precision on waterproofness.
REQ_182	Modified: Exception added.
REQ_183	Deleted: (not applicable in 2027).
REQ_184	Modified: Precisions on Part Number wording and note added.
REQ_185	Modified: Precision on the positioning of the GPS tracker.
REQ_186	Modified: hydrofoil's explicit prohibition.
REQ_188	Modified: clarification that the 25kW limit includes peak power.
REQ_192	Added: Hulls protection.
REQ_193	Added: Only LFP batteries will be allowed from August 2028.
REQ_194	Added: Motor seat structure resistance requirement.
Annex IV	Modified: part number and Datasheet modified.
Rule	Change
7.7	Added: Recommendation on transparent window.

5. Requirements

5.1. The Hull

ENERGY_REQ_1 v1.0

Teams shall use the hulls design provided by the Monaco Energy Boat Challenge organisation.

Note: The design of the hulls is described in Annex I and II. Teams can use the hulls provided by the Organiser or make their own hulls but must stick to the hulls design.

ENERGY_REQ_2 v1.2

Hulls built by the teams shall be:

- 65 kg with beams,
- +/- 2 mm at any measure to the hulls design provided,
- External skin: 500g/m²,
- Internal skin: 300g/m²,
- Sandwich foam: 10mm foam of 100kg/m³,
- At least one horizontal brace and five bulkheads,
- 200 litres of buoyancy.

Note 1: This will be verified by the technical committee thanks to a core drill and a jig.

Note 2: Alterations may be approved by the Organiser, provided they do not provide a competitive advantage and do not modify the hull geometry or colour scheme. Acceptance may be conditioned on technical adjustments determined by the Organiser to ensure competitive fairness. For example, the use of heavier materials may be permitted but overall weight excepting the hulls may be adjusted.

ENERGY_REQ_3 v1.0

Teams shall not modify the hulls and beams provided by the Organiser.

Note: at any time during construction, testing or the Event.

ENERGY_REQ_4 v1.0

Teams shall restore the hulls in the same condition as they were provided by the Organiser.

ENERGY_REQ_192 v1.0

When off the water, the hulls shall be adequately protected.

5.2. Energy source and energy conversion

ENERGY_REQ_5 v1.0

Boats shall not use the following energy sources: ammoniac, fossil fuel, nuclear energy.

Note: As example, boats can use one or more energy of the following list: batteries, hydrogen, kinetic energy, heat, solar, wind, pressured air.

ENERGY_REQ_6 v1.0

Boats shall not use technologies having direct CO₂ emissions.

ENERGY_REQ_7 v1.3

At any moment, the maximum energy stored by a boat shall remain under 10kWh according to the following calculation.

The total energy stored is defined by:

$$E_{tot} = \sum E_i \times f_i$$

With: E_i the energy stored by the source

f_i the energy factor defined by the following table

Energy source (i)	Energy factor (f_i)
Hydrogen	0.35
Battery	1.0
Flywheel	0.5
Heat	0.3
Compressed air	0.2

Note 1: Energy capacity per battery is defined with:

quantity of cells × cell nominal voltage × cell nominal current

Team is entitled to provide the datasheet of cells even if it is not mentioned on the battery datasheet.

Only nominal/typical values will be considered (not minimal / maximal)

Note 2: The calculation of the amount of energy onboard will have to be submitted to the Technical Committee before the Event as per Notice of Challenge.

Note 3: It is allowed to have secondary batteries to power visualisation, communication, satellite positioning and telemetry equipment. This battery is not considered in the calculation of current requirement.

Note 4: Compliance with this requirement is mandatory. Failure to comply will result in disqualification.

ENERGY_REQ_24 v1.0

The energy source shall be securely fixed onboard.

ENERGY_REQ_25 v2.0

Energy container shall be placed at least 500 mm away from the Pilot.

Note: Including secondary energy container.

ENERGY_REQ_26 v1.0

Energy container shall not be located inside the area where the Pilot is seated.

ENERGY_REQ_27 v1.0

Any ventilation from the energy source shall be orientated to blow away from the Pilot.

ENERGY_REQ_30 v1.0

Teams shall not replace or refill the primary energy source once a boat has started a race, except with secondary energy source.

ENERGY_REQ_31 v1.0

Boat shall not use human powered mechanical energy.

Flywheel energy storage

ENERGY_REQ_9 v1.0

Flywheel shall not exceed a weight of 100kg.

ENERGY_REQ_10 v1.0

Flywheel shall not exceed 10 000 rotations per minute.

ENERGY_REQ_11 v1.0

Boat shall embed a mean to measure the rotation speed with a precision of 10rpm at least.

ENERGY_REQ_12 v1.0

Flywheel shall not exceed an external diameter of 2.5 meters.

ENERGY_REQ_13 v1.0

Flywheel housing shall be at least IP20.

ENERGY_REQ_14 v1.0

Flywheel shall have a protective housing capable of containing all released components in the case of disintegration of the system at full power.

Note: The flywheel should not compromise the boat stability.

Heat energy storage

ENERGY_REQ_15 v1.0

Boat's storage medium temperature shall not exceed 500°C.

ENERGY_REQ_16 v1.0

Boat shall embed mean to measure the temperature of the storage medium with a precision of 5°C at least.

ENERGY_REQ_17 v1.0

Teams shall provide the detailed composition of the storage medium.

ENERGY_REQ_18 v1.0

The storage medium shall not leak from the heat energy storage.

ENERGY_REQ_93 v1.1

Any reachable surface shall not exceed a temperature of 60°C.

Gas storage

ENERGY_REQ_8 v2.0

Pressure storage shall have at least one analogical manometer.

ENERGY_REQ_94 v1.0

Onboard gas pressure shall not exceed 700 bars.

ENERGY_REQ_19 v2.0

Gas pressure system shall integrate pressure relief devices immediately downstream any pressure reducer or compressor.

ENERGY_REQ_21 v1.0

In case of explosive gas, the pressure relief valve shall be collected to the vent.

Note: Refer to H2 safety part for vent requirements.

ENERGY_REQ_22 v1.0

Statement of testing and service for all gas pressure tanks proving compliance with the manufacturer recommendations shall be provided to the Technical committee.

ENERGY_REQ_123 v1.0

Gas cylinders shall conform to following standards: EC79¹, TPED² or R134³.

Note: All other means of compliancy shall be submitted to the Technical Committee at registration stage.

Batteries

ENERGY_REQ_23 v1.0

Batteries shall be used within their nominal capacities according to their datasheet.

Note: in case of homemade battery, the team shall provide a datasheet.

ENERGY_REQ_155 v1.2

If the battery is located inside a container, the container shall feature an overpressure certified and adequately sized valve oriented toward the back of the boat.

Note: If a battery is enclosed within multiple nested containers, each container shall be fitted with an appropriate overpressure relief valve.

ENERGY_REQ_56 v2.1

If the battery is located inside a container, the container shall feature a Fire Extinguisher Port located on the port side of the container, in order to be accessible for external personnel.

ENERGY_REQ_189 v1.0

If the battery is located inside a container, the container shall provide an external charging interface allowing the battery to be recharged without opening the container.

ENERGY_REQ_171 v1.0

Every battery with a capacity superior to 100 Wh and all containers in which they are located shall figure the W026 symbol form ISO 7010:2019.

Note: the symbol size shall be no less than 10 cm, or the battery or container size.

ENERGY_REQ_172 v1.0

Every battery with a capacity superior to 100 Wh and all containers in which they are shall figure the capacity of the battery.



Figure 1
W026 symbol
from ISO 7010:2019

¹ Regulation (EC) No 79/2009

² Transportable Pressure Equipment Directive 2010/35/EU

³ Regulation No 134 of the Economic Commission for Europe of the United Nations (UN/ECE) — Uniform provisions concerning the approval of motor vehicles and their components with regard to the safety-related performance of hydrogen-fuelled vehicles.

Note: the letters size shall be no less than 2 cm, or the battery or container size.

ENERGY_REQ_173 v1.0

Every battery with a capacity superior to 100 Wh and all containers in which they are shall figure the battery composition.

Note: the letters size shall be no less than 2 cm, or the battery or container size.

ENERGY_REQ_193 v1.0

From August 2028, lithium-ion batteries shall use LFP chemistry. Solid-state batteries are also permitted. Any other battery technology shall require prior approval from the Technical Committee.

Solar panels

ENERGY_REQ_28 v2.0

Solar panels shall have a surface inferior or equal to 4m² including the mounting frame.

Sails

ENERGY_REQ_29 v2.0

The maximum size of the sail for a boat shall be 20m².

Note: Team will have to demonstrate the boat is not able to capsize in all conditions.

5.3. The Engine, Propulsion & Steering

ENERGY_REQ_32 v1.0

Boat shall be able to move and be manoeuvrable in forward and reverse gear.

Note: manoeuvrability capacities are appreciated by the Technical committee during sea certification test.

ENERGY_REQ_154 v1.0

The steering system or shaft shall rotate of at least 40° on each side (port and starboard).

ENERGY_REQ_33 v1.0

The steering gear of the boat shall be of a size for adequate controllability.

ENERGY_REQ_34 v1.0

Steering gear shall operate smoothly.

ENERGY_REQ_35 v1.0

Steering gear shall be free of play both in loaded and unloaded conditions.

ENERGY_REQ_36 v1.0

Steering gear shall be capable of being operated without power from the Pilot seat without any modification to the system.

ENERGY_REQ_37 v1.0

Boat shall sail at a speed of at least 3 knots.

ENERGY_REQ_188 v1.1

The sum of instantaneous power consumption of all motors shall not exceed 25 kW.

Note: No power exceeding 25 kW will be allowed, even during a peak.

5.4. The Cockpit

ENERGY_REQ_38 v2.1

The cockpit must be fastened to the beams with at least two clamps or jaws per beam, continuously enveloping the entire circumference of the beam, over a minimum width of 50mm per clamp/jaw installed on the beams symmetrically on either side of the ship's centreline with a minimum spacing of 750 mm. These clamps must be fitted with a rubber gasket between the beam and the clamp with a minimum thickness of 1mm.

ENERGY_REQ_39 v1.0

Every component of the cockpit or in the cockpit shall be fixed.

ENERGY_REQ_194 v1.0

The motor seat structure shall withstand a torque load equivalent to at least 200% of the maximum torque generated by the motor at full power.

Note: The design should integrate peak power to full power.

ENERGY_REQ_41 v1.0

The driving position shall be such as the feet of the Pilot are positioned in front of his body and be free of any potential hazards.

ENERGY_REQ_42 v1.0

Seats shall include a headrest.

ENERGY_REQ_43 v1.0

Seats shall be designed in such a way that the Pilot has a clear field of view and have unobstructed hearing at all times.

ENERGY_REQ_44 v1.0

Boat shall be designed in such a way that the Pilot is able to evacuate the boat within 5 seconds without any form of outside help.

Note: This will be demonstrated by means of an evacuation test.

ENERGY_REQ_45 v1.0

The cockpit shall be designed in such a way that the Pilot is not fully enclosed.

Note: Hatches that need to be opened before the Pilot can evacuate the boat are not allowed.

ENERGY_REQ_46 v1.0

Boat shall be designed in such a way that the Pilot is not restrained by any means to the boat.

Note: Safety belts are also not allowed.

ENERGY_REQ_47 v1.0

The cockpit shall be self-draining.

ENERGY_REQ_48 v3.1

The overall weight excluding the hulls shall not exceed 250kg.

Note: The boat will be weighted with the hulls and the pilot.

Note 2: The hulls provided by the Organiser are supposed to be $65 \pm 1\text{kg}$.

Note 3: Overweighted boat won't be allowed to enter the water.

ENERGY_REQ_49 v2.0

The cockpit shall be fitted with high visibility tape to reflect the sunlight:

- 500cm² on starboard side,
- 500cm² on port side,
- 250cm² on frontward,
- 250cm² on backward.

ENERGY_REQ_186 v1.1

Hydrofoils are not permitted. No cockpit appendices shall be in contact with the water, except for the motor and the water pump.

Bulkhead

ENERGY_REQ_50 v1.1

A bulkhead shall safely isolate the boat's propulsion and energy storage systems from the driver's compartment.

ENERGY_REQ_51 v2.0

The bulkhead shall be at least A1 fire resistant according to Euroclass.

ENERGY_REQ_52 v2.0

A bulkhead shall protect the pilot from an explosion of the energy storage in any directions.

ENERGY_REQ_54 v1.0

Energy equipment that might generate a risk for the Pilot shall be placed outside the driver's compartment behind a bulkhead.

ENERGY_REQ_55 v1.0

If holes are made in the bulkhead to pass through wires or cables, they shall be protected by a grommet or similar protective material to prevent chafing or damage.

5.5. *Electrics/Electronics*

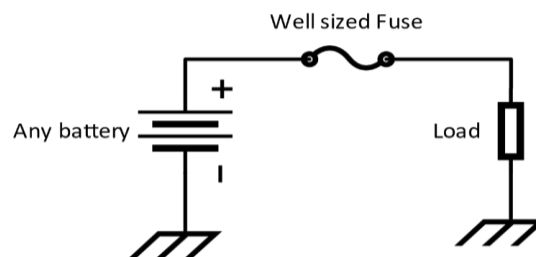
ENERGY_REQ_57 v1.0

Boat shall integrate well sized wire gauge.

ENERGY_REQ_58 v1.1

Boat electric system shall integrate well sized fuse just after the energy storage.

Note: Including telemetry batteries.



ENERGY_REQ_59 v1.0

The continuous current rating of the overcurrent protection shall not be greater than the continuous current rating of any electrical component.

Note: for example, wire, or other conductor that it protects.

ENERGY_REQ_60 v1.1

All live parts of the electrical system shall be protected from being touched.

Note: this is tested with a 100mm long, 6mm diameter insulated test probe when the boat is ready to sail.

ENERGY_REQ_61 v1.0

All wiring that runs outside of the electrical system enclosure shall be of orange colour.

Note: either with orange tape or the cable itself is orange.

ENERGY_REQ_62 v1.0

Electronics shall be housed in IP56 watertight compartment and should be cooled.

ENERGY_REQ_63 v1.0

Electrical circuits with more than 6V AC or 15V DC shall be at least IP2X and insulated from other circuits.

Note: Busbar included.

ENERGY_REQ_187 v1.0

If a team intends to use a voltage above 100 V, it shall obtain approval from the Technical Committee during the registration phase.

ENERGY_REQ_191 v1.0

A training was attended by all of those who will work on high voltage systems (100 V or higher). This certification can be issued by any training centres, or by the university. The validity of the certification will be checked during the technical inspections.

5.6. Monitoring

Internal

ENERGY_REQ_67 v2.0

The temperature of all batteries (incl. telemetry) shall be monitored. Temperature monitoring system shall provide the temperature to the Pilot.

ENERGY_REQ_68 v2.0

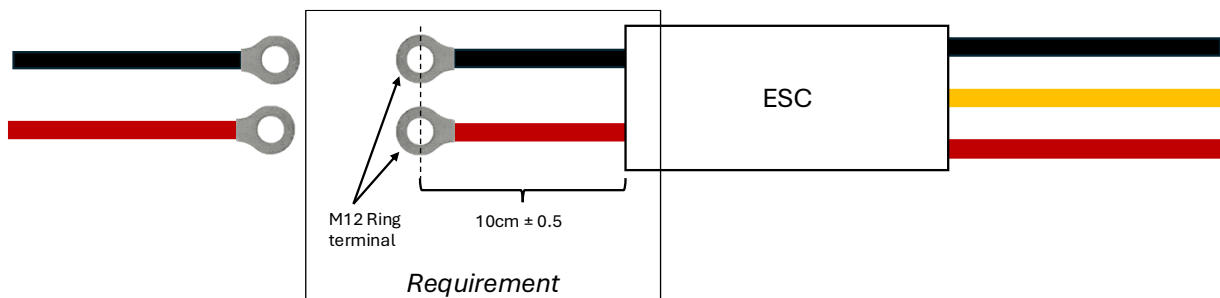
Temperature monitoring system shall warn the pilot when 90 % of maximum monitored temperature is reached.

Note: 90 % of the maximum temperature of the component in Celsius degrees.

External

ENERGY_REQ_181 v1.2

The input power to each ESC shall comply with the following diagram to enable integration of the sensing device system.



Note 1: Power sensors provided by the competition are not waterproof and have to be protected from humidity and water at any moment.

Note 2: Assembly of the power sensors provided by the competition has to be done using 2 nuts per bolt strongly screwed or vibration resistant system.

ENERGY_REQ_195 v1.0

The team shall provide a free mounting volume of 220 × 111 × 80 mm (L x W x H) for each power sensors device provided by the Organiser. The Power sensors device are the one corresponding to ENERGY_REQ_181

- This area shall be water proof
- Each traction chain of the boat has to be monitored. In case of multiple traction chain (distinct motors and battery/energy sources), one sensor has to be place per chain and therefore one mounting volume of 220 × 111 × 80 mm (L x W x H) has to be provided per chain.
- Schematics of the Power Sensor device is provided in Annex V.

ENERGY_REQ_182 v2.0

Each battery *shall have a pass-through hole of at least 3.2 mm diameter* to allow installation of a temperature sensor by the Organiser. It shall be possible to position the temperature sensor *in contact with the battery cells* at the core of the battery.

Exception is made for batteries that are inside a watertight casing or that are provided by manufacturer within a non-accessible casing. In this case the temperature sensor shall be fixed on the external wall of the battery container, as close as possible from cells, protected from direct water contact and protected from direct sun heating and outside temperature variation.

Note 1: If the team is not permitted to modify its COTS battery pack, it shall contact the Technical Committee at the registration stage.

Note 2: Temperature sensors provided by the competition are not waterproof, they have to be protected from water at any moment.

ENERGY_REQ_184 v1.1

The team shall provide a power interface of 10 – 28 V, maximum 35 W.

The power interface connector shall be a Part Number :

- AD-02BFFA-LL7001
- 1AD-02BFFM-LL7A**XX**
- AAD-02BFFM-LL7A**XX**
- AAD-02AFFM-LL7A**XX**, (**XX is the cable length from 01 to 99meters**)

from Amphenol LTW, with the positive input on Pin 1 and Ground on Pin 2. As shown on ANNEX IV.

Note: 35 W is a maximum, nominal will be communicated further.

ENERGY_REQ_185 v1.1

The team shall provide a free mounting volume of $190 \times 150 \times 122$ mm (L x W x H) for an additional monitoring device from the Organiser.

This area shall be:

- open to the sky,
- at least 40 cm above the beams of the hulls, outside all the closed compartments,
- placed laterally less than 40 cm away from the longitudinal axis of the boat (fig. 2b) as the x-y (L x W) plan has to be as parallel as possible to the navigation boat trim.

Note 1: Sensor cabling from the monitoring device to the various sensors will be installed by the Organiser.

Note 2: Teams may anticipate the wiring path to minimizing externally visible cables and must maintain a minimum distance of 10 cm from all power cables at all times.

Note 3: Crossing above a power cable is permitted only at a perpendicular angle.

Note 4: A camera will be installed on each boat by the Organiser, with the fixation method to be defined.



Figure 2a
XYZ axis orientation
(L x W x H)

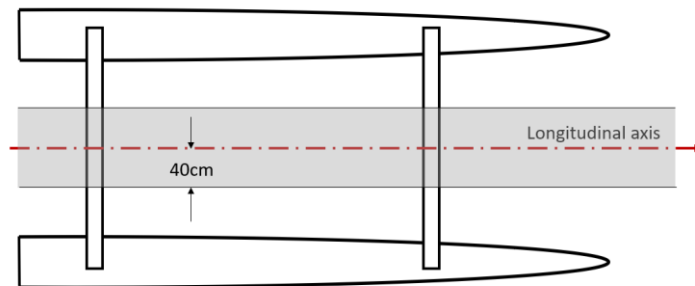


Figure 2b
Schematic view of the boat's longitudinal axis

5.7. Kill cord & Emergency stop push button

NOTE: pictures used for clarity

Kill cord

ENERGY_REQ_70 v1.0

Boat shall integrate a kill cord.

ENERGY_REQ_71 v1.1

Kill cord shall cut the power supply to the engine instantly, less than 1 second, as soon as the Pilot loses control of the boat or when the Pilot leaves the boat, whether voluntarily or involuntarily.

ENERGY_REQ_72 v1.1

One end of the kill cord shall be always secured to the pilot's clothes when sailing and the other end to a device onboard set to cut off the power/ignition of the engine if the kill cord is pulled.

ENERGY_REQ_73 v1.0

The kill cord shall be clearly visible at 3 meters distance.

ENERGY_REQ_74 v1.1

When the cord is replaced, at least one more additional action shall be taken to enable the engine to start running again.

Note: waiting is not considered as an action.



Figure 3 Kill cord example

Emergency stop push button

ENERGY_REQ_75 v1.0

Boat shall be fitted with an emergency stop push button.

Note: kill switch with "mushroom" head. Figure 4 is an example.

ENERGY_REQ_76 v2.0

Emergency button shall be clearly marked and visible thanks to the following colours: button in red over a 90 mm yellow circle.



Figure 4
Emergency stop
button example

ENERGY_REQ_77 v2.0

The button shall have a diameter of at least 30mm.

ENERGY_REQ_78 v1.1

The international electrical symbol consisting of a red spark on a white-edged blue triangle shall be affixed in close proximity to the emergency stop button and its height shall be minimum 100mm.



Figure 5
Emergency stop
symbol

ENERGY_REQ_79 v1.2

When activated, the emergency stop button shall isolate all power sources from the boat. Telemetry power sources are included in this requirement.

Note: Relays are allowed but no software stop.

ENERGY_REQ_80 v1.1

When emergency stop button is deactivated, at least one more additional action shall be taken to enable the engine to start running again.

Note: waiting is not considered as an action.

ENERGY_REQ_81 v2.0

Emergency stop button shall be located on the starboard side and located in front of the pilot.

ENERGY_REQ_82 v1.0

Emergency stop button location shall be so that the emergency stop button can be activated by the Pilot from its seating position.

ENERGY_REQ_83 v1.0

Emergency stop button activation shall not delay the evacuation.

ENERGY_REQ_84 v1.0

Emergency stop button shall be accessible by external personnel.

ENERGY_REQ_85 v1.0

Emergency stop button shall be located within one meter of the starboard side.

ENERGY_REQ_86 v1.0

Emergency stop button shall be oriented toward outside.

Note: Emergency stop button oriented up won't be considered compliant.

ENERGY_REQ_87 v1.0

Emergency button shall remain functional at all times while the Pilot is onboard.

ENERGY_REQ_88 v1.0

Emergency button shall be properly fixed to boat.

ENERGY_REQ_89 v1.0

When activated by a hit, the emergency button shall remain stable.

5.8. Safety & security Equipment

ENERGY_REQ_90 v1.0

The boat shall not have sharp edges.

ENERGY_REQ_91 v1.0

When off the water, the propellers shall be adequately protected or removed.

ENERGY_REQ_92 v1.0

All rotating components in or on the boat shall be adequately shielded.

Note 1: to prevent unintentional contact while on and off the water.

Note 2: when in the water, propellers used for propulsion are exempted from this rule.

Safety equipment

ENERGY_REQ_95 v1.1

Safety equipment shall be always reachable and functional to the Pilot at all times.

ENERGY_REQ_96 v1.0

Safety equipment shall provide a towing floating bridle of at least 10mm diameter.

ENERGY_REQ_97 v1.0

The towing floating bridle shall be fitted to the boat.

ENERGY_REQ_99 v1.0

Safety equipment shall provide a floating towline of at least 10 meters and a minimum diameter of 10mm.

ENERGY_REQ_98 v1.1

The floating towline shall be accessible to the Pilot.

ENERGY_REQ_100 v1.0

The floating towline shall be secured to the bridle.

ENERGY_REQ_101 v1.0

Safety equipment shall provide a paddle with a minimum overall length of 100cm, a minimum blade length of 30 cm and a minimum blade width of 13 cm.

ENERGY_REQ_102 v1.0

Safety equipment shall provide a boat hook with a minimum length of 100cm.

Note: it may be combined with the paddle. However, a regular paddle handle is will not be considered a hook. A hook needs to be able to catch.

ENERGY_REQ_103 v1.0

Safety equipment shall provide a uniformly coloured orange or red warning flag with a minimum size of 30 cm x 30 cm attached to a stick or similar structure with a minimum handle length of 100 cm.

ENERGY_REQ_104 v1.0

The orange/red flag shall not be secured to the paddle or the boat hook.

ENERGY_REQ_105 v1.0

Safety equipment shall provide an audible warning system.

Note: such as a ship's horn audible at reasonable distance, such as an orally operated horn, an air horn or an electrically operated horn, a whistle fixed to the lifejacket will also be accepted as audible warning system.

Extinguisher

ENERGY_REQ_106 v1.0

There shall be onboard approved fire extinguisher with a minimum capacity of 1 kg of extinguishing material suitable for extinguishing fires on board.

ENERGY_REQ_107 v1.0

The fire extinguisher shall be ABC type.

ENERGY_REQ_108 v1.0

Fire extinguishers shall show a valid approval.

ENERGY_REQ_109 v1.0

The approval should show the date the fire extinguisher was tested last and the date when the next test is due.

ENERGY_REQ_110 v1.0

The fire extinguisher shall be mounted in a position such that it can be reached easily by the Pilot from their normal seating position in the cockpit.

Note: It can be fitted under, in front aside of the Pilot, but can't be fitted in the back of his seat.

ENERGY_REQ_111 v1.0

The fire extinguisher shall be prevented from being dropped into the water after taking it out of its housing attachment.

Hydrogen safety

ENERGY_REQ_156 v1.0

There shall be a dedicated bulkhead isolating the gas storage from electrical components.

ENERGY_REQ_64 v2.0

Hydrogen boats shall have the cockpit grounded at all times.

Hydrogen Piping and Storage

ENERGY_REQ_157 v1.0

The Team shall justify hydrogen compatibility if other materials than stainless steel 316L are used for hydrogen piping.

ENERGY_REQ_158 v1.0

Piping subjected to a pressure greater than 10 bar shall be either:

- Rigid metallic piping,
- Flexible hose fitted with steel anti-whipping cable securely fastened every 15cm to an adequate structural element.

ENERGY_REQ_159 v1.0

Piping subjected to a pressure greater than 10 bar shall not be located closer than 1m to the Pilot.

ENERGY_REQ_122 v1.0

A manual shutoff valve shall be placed between the hydrogen storage and the hydrogen consumer, closest to the storage.

ENERGY_REQ_124 v1.0

Gas cylinders shall be of proprietary manufacture, designed and built for the pressure being used, certified and labelled or stamped appropriately.

ENERGY_REQ_125 v1.0

Gas cylinders shall be used within the temperature range specified by the manufacturer.

ENERGY_REQ_126 v1.0

The axis of gas cylinders shall not point at the driver.

ENERGY_REQ_127 v2.0

Gas cylinders shall be insulated from any heat sources, including direct sunlight.

ENERGY_REQ_128 v1.0

Gas cylinders onboard and stored onshore shall be fitted with a gas cylinder gripping tulip strong enough to prevent any mechanical hazard in case of 1000N force applied in all axes.

ENERGY_REQ_129 v1.0

All used parts shall be appropriate for the maximum possible operating pressure.

ENERGY_REQ_130 v1.0

Hydrogen metallic piping and parts shall be earthed.

ENERGY_REQ_131 v1.0

A leak test shall be performed by the Technical Committee at the first time the hydrogen line is put under pressure.

ENERGY_REQ_132 v1.0

Leak tests shall regularly be performed on the hydrogen line.

Hydrogen Inside Enclosed Compartments

ENERGY_REQ_160 v1.0

Compartments containing hydrogen shall feature an automatic isolation device, located outside the compartment and isolating all the piping of the compartment from any hydrogen source.

ENERGY_REQ_161 v1.0

Compartments containing hydrogen shall feature a hydrogen sensor.

ENERGY_REQ_115 v2.0

In the event of a leak detection, automatic safety actions shall be the following:

- Automatic isolation devices shall be immediately closed
- All non-Atex electrical components inside the compartment shall be immediately shut down
- Ventilation of the hazardous area shall be ensured for at least 60s after the leak detection

Note: hazardous area can be defined according to IEC60079-10-1:2020, or a justification can be submitted to the Technical Committee.

ENERGY_REQ_162 v1.0

Compartments containing hydrogen shall be ventilated.

ENERGY_REQ_163 v1.0

The ventilation shall ensure to remain below 50% LEL (2% H₂ in air) in case of probable leak scenario. The team shall justify the ventilation ratio by calculation taking into account pressure drops.

Note 1: A probable leak scenario is considered as follows:

- For piping equal to or below ½" diameter: 0.1mm²
- For piping bigger than ½" diameter: 0.1xD mm²

Note 2: In case of natural ventilation, a 3 knots wind speed can be considered for an inlet oriented towards the front of the boat.

ENERGY_REQ_164 v1.0

In case of natural ventilation, the inlet shall be located below any hydrogen piping to ensure no gas can come out through it.

ENERGY_REQ_165 v1.0

In case of natural ventilation, the outlet shall be located 50cm above the driver's head

ENERGY_REQ_166 v1.0

In case of forced ventilation, a fan located inside the hydrogen flow in case of leak shall be certified for ATEX zone 2 with hydrogen gas and non-sparking.

Note: an example of suitable certification is II2G Ex IIC nA T6 Gb.

ENERGY_REQ_167 v1.0

In case of forced ventilation, a fan located outside the hydrogen flow in case of leak does not require ATEX zone 2 certification. If the fan is located above hydrogen piping, the Team shall justify that no hydrogen will reach it in case of leak.

ENERGY_REQ_168 v1.0

In case of forced ventilation, the outlet shall be located 50cm away from the driver.

ENERGY_REQ_169 v1.0

In case of forced ventilation, the outlet shall be oriented towards the back of the boat.

8.1.1. Hydrogen release to the air

ENERGY_REQ_117 v1.0

Hydrogen releases at the fuel cell air exhaust shall be collected and rejected toward the back of the boat and at least 1.5m away from the Pilot.

ENERGY_REQ_118 v1.0

All expected releases of hydrogen shall be collected and rejected to the vent.

Note: it includes the pressure release valve, and all other expected hydrogen releases. Fuel cell air exhaust is not considered.

ENERGY_REQ_119 v1.0

The hydrogen vent shall be oriented upward.

ENERGY_REQ_120 v1.0

The release point of the vent shall be at least 1.5m above the hulls.

ENERGY_REQ_121 v1.0

The vent shall not be used for the flag.

5.9. Pilot Requirements and Communication

ENERGY_REQ_135 v1.0

The weight of the Pilot ready to sail, meaning with overalls, helmet, lifejacket, shoes and communication system, shall be at least 60 kg otherwise, the boat will be ballasted to reach 60kg.

Note: In case there is more than 1 Pilot, the lightest Pilot shall be used to calculate the correct amount of ballast to be put onboard.

ENERGY_REQ_136 v1.0

Ballast, if used, shall be properly fixed to the boat, in or near the Pilot's seat and accessible for inspection.

ENERGY_REQ_137 v1.0

Pilot on board the boat shall wear an approved lifejacket of 100 Newton buoyancy capacity or equivalent.

ENERGY_REQ_138 v1.0

If the lifejacket is not of the rigid type, the lifejacket shall be automatically inflatable when coming into contact with water.

ENERGY_REQ_139 v1.0

In case of an automatic lifejacket, the due date of the device shall be after the Event.

ENERGY_REQ_140 v1.0

Pilot shall wear an open face, bright colour fluorescent helmet, a bright colour flame resistant clothes or boiler suit and non-buoyant shoes.

ENERGY_REQ_190 v1.0

Pilot must wear tear resistant gloves.

Note: for example EN 388:2016

Communication

ENERGY_REQ_141 v1.1

Pilot shall be capable of communicating to one member of their onshore team.
VHF is forbidden.

ENERGY_REQ_142 v1.0

The mean of communication shall have a broadcast range of at least 2 nautical miles.

ENERGY_REQ_143 v1.0

The mean of communication shall be integrated into the helmet of the Pilot.

ENERGY_REQ_144 v1.0

The mean of communication shall be waterproof.

ENERGY_REQ_145 v1.0

The mean of communication shall be supplied with a battery of sufficient capacity to last at least one race without being recharged.

5.10. *Appearance of the Boats*

ENERGY_REQ_146 v1.0

The first third of the hull forward shall be reserved for branding or markings provided by the Organisation.

Note: The rest of the hull and the cockpit is free for team's markings.

ENERGY_REQ_147 v1.0

Markings on the hulls or the cockpit shall not be in conflict with sound moral standards whatsoever.

Note 1: The Organisation reserves the right to refuse a boat that displays marking judged contrary to the values and the messages of the Yacht Club de Monaco.

Note 2: A boat will not be authorised to start a race if the markings have not been complied with.

Note 3: The Organisation reserves the right to mount a camera on each boat.

ENERGY_REQ_170 v1.0

The cockpit must be built on the state of the art of the boating industry (professional appearance), the design has to be approved by the Design Jury at validation of projects stage in January as per Notice of Challenge.

ENERGY_REQ_148 v1.0

Boat shall have the appropriate symbols from ISO 7010:2019 on a visible place of the cockpit.

Note: symbols can be found on https://en.wikipedia.org/wiki/ISO_7010.

ENERGY_REQ_149 v1.0

Boat shall fly a flag of his/her organisation nationality.

ENERGY_REQ_150 v1.0

The flag shall fly at least 2m above the water.

ENERGY_REQ_151 v1.0

The flag shall have a minimum width of 30cm.

ENERGY_REQ_152 v1.0

All markings on the hull shall be removed by the end of the Event.

6. Inspections / Racing Test

- 6.1. The Organisation is entitled to conduct inspections of the boats at any time of its own choosing. The participants are bound to cooperate with such inspections.
- 6.2. The Organiser will inspect all boats for full compliance with the Technical Rules prior to racing. During the inspection the participants are required to present their boat in a race-ready condition.
- 6.3. During the inspections, the participants are required to demonstrate the racing performance of their boats. During this demonstration, the participants must race a prescribed circuit. The boat and the Pilot will be judged on the following aspects:
 - Controllability of the boat,
 - Racing skills of the Pilot,
 - Freeboard in racing condition.
- 6.4. Boats that fail to comply with the applicable Technical Rules or the racing test (final approval) will not be allowed to enter a race.
- 6.5. During the Event the boats are not allowed to leave the paddock without permission. Leaving the paddock without the prior permission of the Organiser will lead to disqualification.
- 6.6. The boat and Pilot weight may be checked at any time by the Technical Committee. Daily random weight checks will be performed.
- 6.7. The energy source can be controlled at any time during the Event by the Technical Committee.
- 6.8. The Technical committee may decide not to authorize a pressure storage if not considered safe.
- 6.9. Technical inspections will have two parts:
 - Onshore test, leading to first security and equity approval.
 - Without this approval, boat is not allowed to enter the water, except if a Technical Committee member agreed and is present,
 - Water test, leading to final approval.
- 6.10. For technical inspection, the team shall provide to the Technical committee at least:
 - Electronic schematics including fuses, emergency stop button and gauge size
 - Every electrical component's datasheet (including cell's if the battery is home-made)
 - Embedded energy calculation
 - A weight distribution diagram
 - A list of every container's composition
 - For boat using hydrogen, a P&ID and the fuel cell datasheet.

The technical committee might ask for any documents it finds useful to assess the boat and current regulation compliancy.

7. Failure & Modifications

- 7.1.** After technical inspection approval, it is prohibited to add or remove any part, except with the written consent of the Technical Committee.
- 7.2.** Propeller may be changed between races.
- 7.3.** The Organiser must be informed of any failure.
- 7.4.** In case of failure onboard, teams are entitled to repair and/or replace the failed or flawed components with identical ones.
- 7.5.** Reparation with non-identical part requires the authorisation of the Technical committee.
- 7.6.** Electronical parts will be sealed by the Technical committee after the technical inspections to prevent any modification. Seals shall not be broken unless technical committee authorization.
- 7.7.** The Technical Committee strongly recommends that the battery enclosure be fitted with a transparent inspection window and with any necessary charging or diagnostic connectors accessible from outside the enclosure. This allows visual inspection and connection without opening the battery enclosure when required.

ANNEX I – Hull Package

The hull is made of 2 hulls and 2 beams bolted together. The beams are round carbon fibre poles. They will be used to support and secure the cockpit.

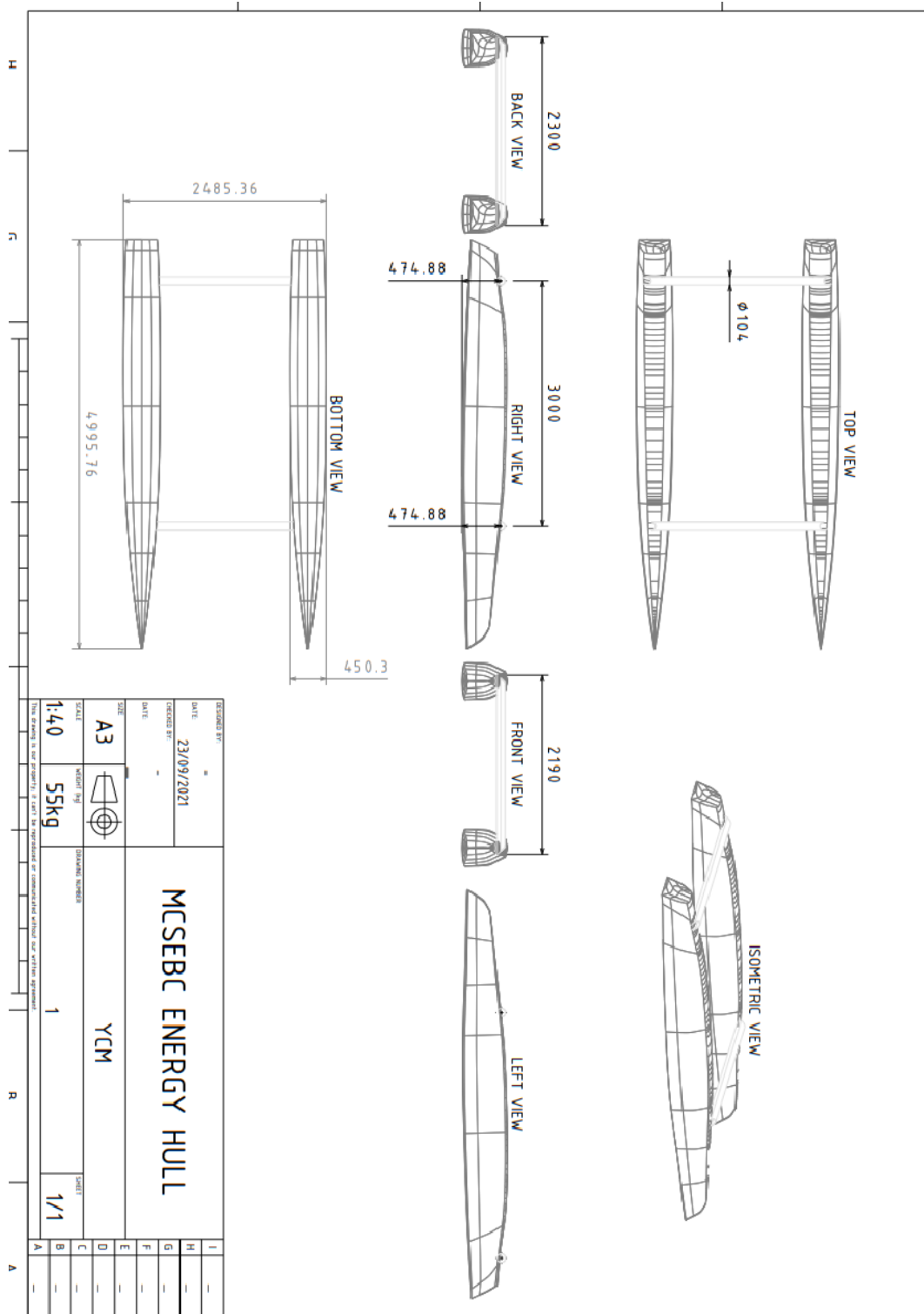
Dimensions

Overall Length	5 m
Overall Width	2.5 m
Free board height	0.45 m
Beam Diameter	10.4 cm
Longitudinal distance between beams	3 m
Total Weight of hulls + beams	65 kg

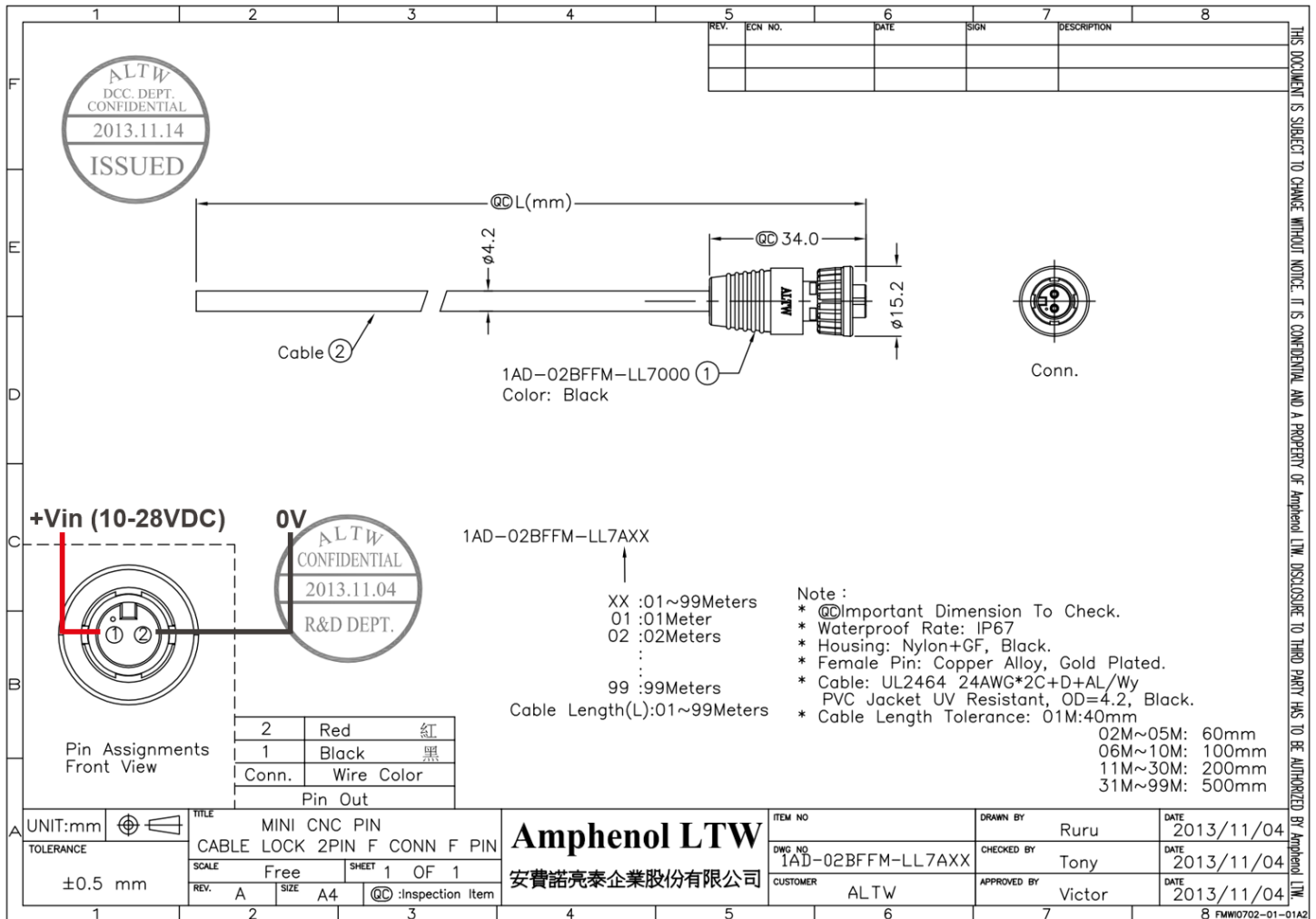
The hull has been designed to comply with French Category C meaning that the hull has been designed to sustain winds up to 6 Beaufort and a 2-metre significant wave height. Hulls have been constructed to be unsinkable. This is achieved by means of watertight compartments and buoyant materials integrated to the hull.

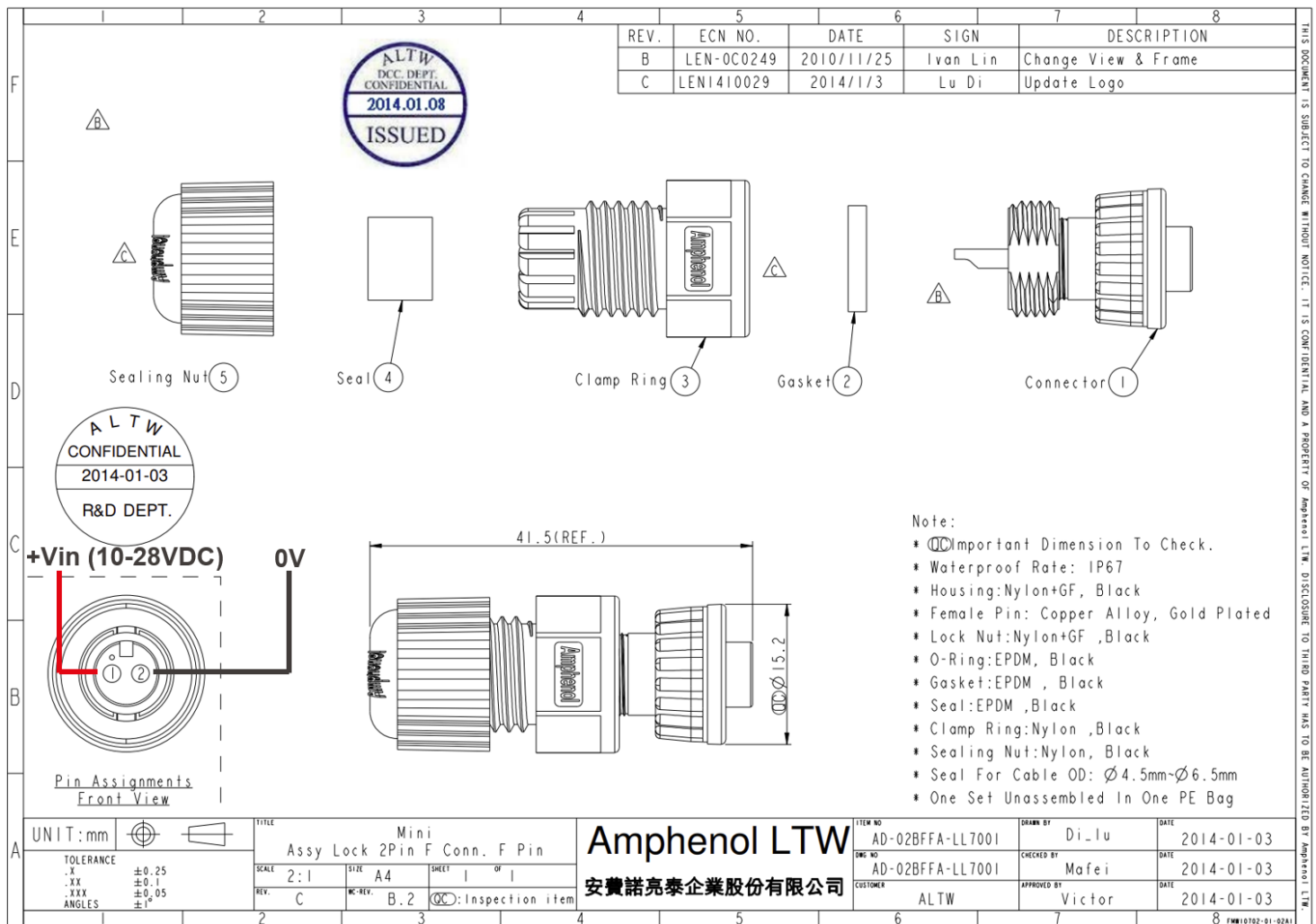
You may download the Hull Package available to all teams on the team's Dashboard, or request from the Organisation.

ANNEX II - Hull Drawings



ANNEX III – External monitoring power supply interface





ANNEX IV – Power Sensor Device schematics

